GAN Plots (1FA006)

Dabbugunta Hyndavi Anand

The Jupyter notebook can be found in this GitHub repository.

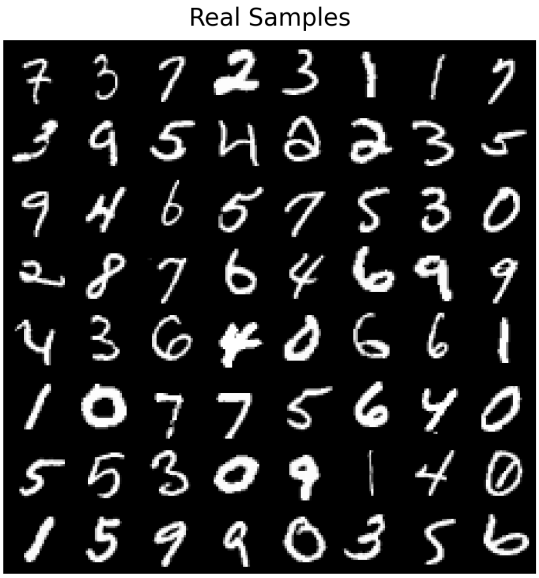

Figure 1: Samples from the MNIST Dataset

Vanilla GAN

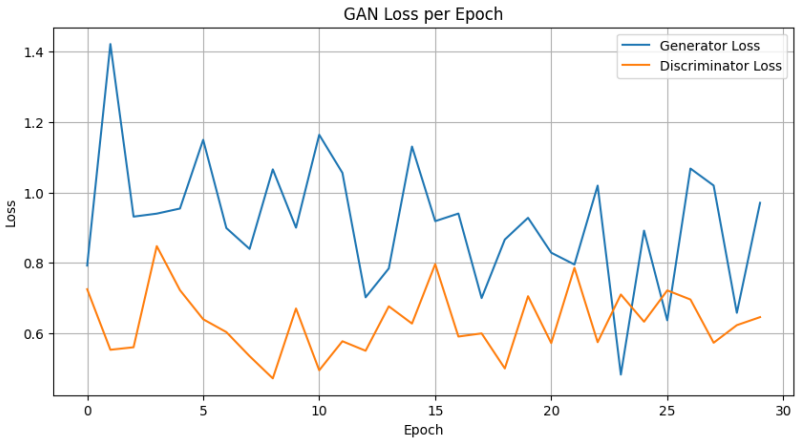

Figure 2: Loss curve for Vanilla GAN. The oscillating losses show that the Generator and Discriminator are competing with each other. This is a sign of a healthy GAN training.

Wasserstein GAN (WGAN)

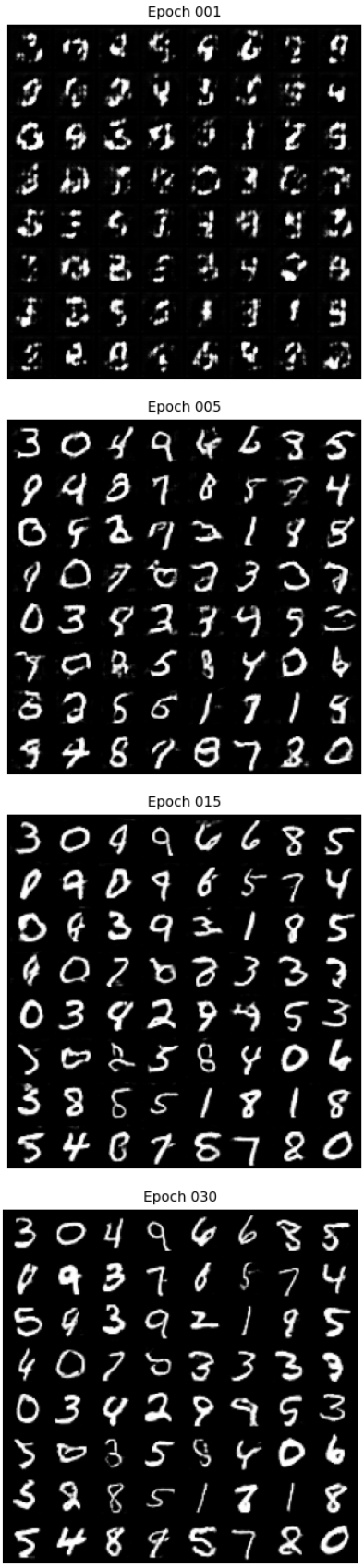

Figure 3: Generated batch images at various points in training for GAN. This tells us the majority of the image is generated in the first 5 epochs and it goes through minor changes in the later epochs. The generated images look like they would belong in the MNIST dataset save for a few samples.

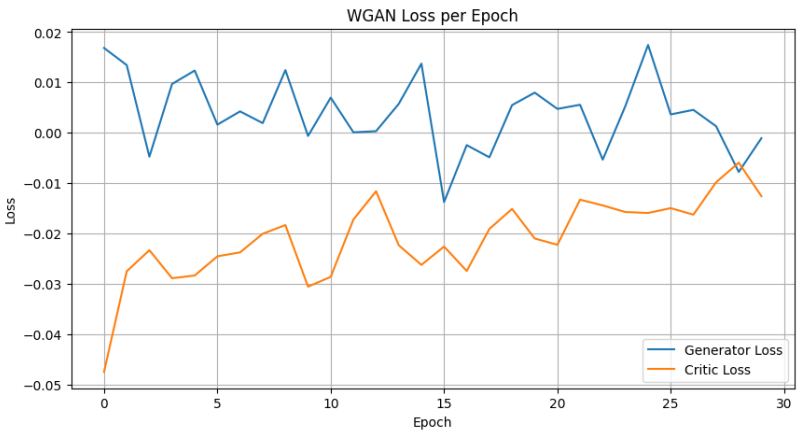

Figure 4: Loss curve for WGAN. Much like the typical GAN, the critic and generator losses oscillate around each other. The critic tries to push real image score up and fake image score down to maximize their difference, while the generator tries to push fake score up to fool the critic. As training progresses both losses trend toward 0, meaning the generator is getting good enough that the critic can't separate real and fake scores by much.

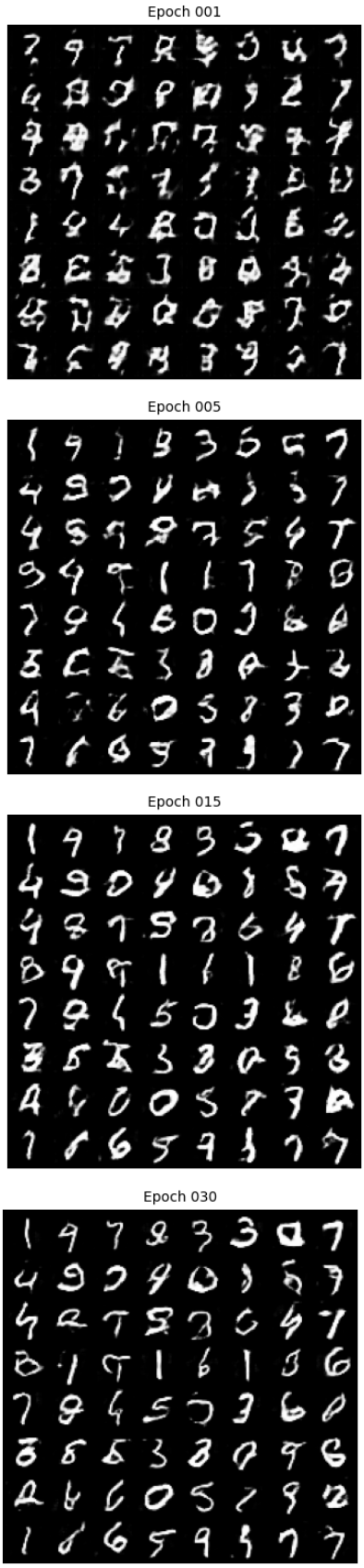

Figure 5: Generated batch images at various points in training for WGAN. These generated images do not look as good as the Vanilla GAN ones but I think this can be improved by using Gradient penalty instead of weight clipping.